Magnetic Ordering in Chiral Nanotubes with Competing Interactions

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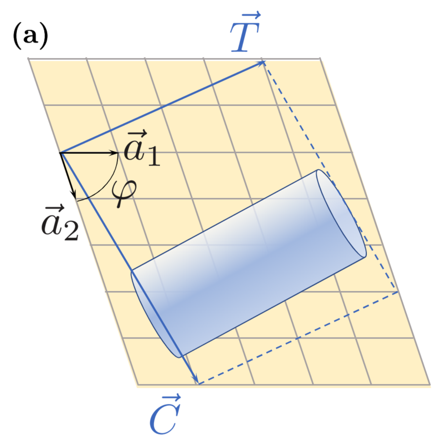
1. Abstract

In this work, we investigate the low-temperature equilibrium magnetic configurations of single-wall magnetic nanotubes with competing exchange and dipolar interactions using Monte Carlo simulations within the framework of the classical 3D Heisenberg model. The nanotubes are constructed by folding planar spin lattices at various chiral angles, allowing control over radius, length, and underlying lattice. Through the analysis of the different magnetization components and energies, we have revealed a rich variety of stable magnetic states, including uniform, vortex-like, and helical configurations, whose nature strongly depends on the tube aspect ratio, length, chirality, and the ratio $\gamma = g/J$. The inclusion of anisotropy further enriches the phase space, stabilizing radial and mixed magnetization textures.

2. Nanotube Geometry

To describe nanotube structure, we use the chiral vector, $\vec{C} = n\vec{a}_1 + m\vec{a}_2$, where the radius is given by: $R = |\vec{C}|/(2\pi)$. Three **tube types**:



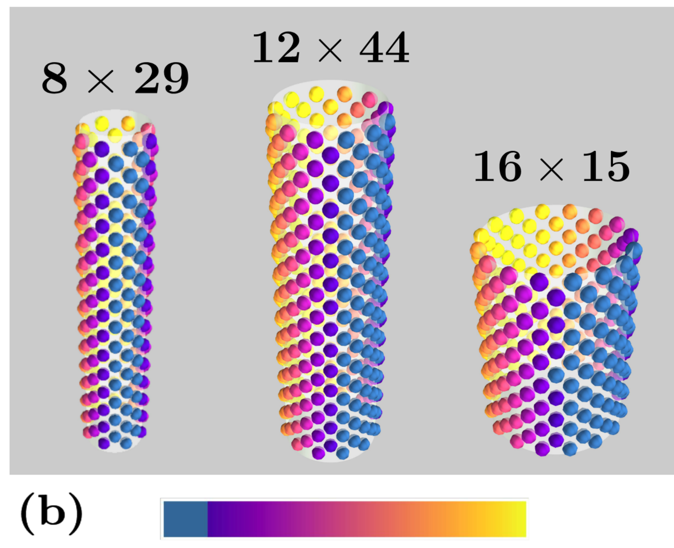
- **AA**: Square lattice ($\phi = \pi/2$, n or $m = 0$)
- **AB**: Triangular lattice ($\phi = \pi/3$, n or $m = 0$)
- **ZZ**: Zigzag ($\phi = \pi/2$, $n = m$)

3. Model and simulation method

We employ **Monte Carlo** simulations using the **Metropolis algorithm** with simulated annealing. The analysis is performed in **cylindrical coordinates**, where the **magnetization components** are:

$$m_\rho = \frac{1}{N}\sum_i s_i^\rho, \quad m_\phi = \frac{1}{N}\sum_i s_i^\phi, \quad m_z = \frac{1}{N}\sum_i s_i^z$$

Starting from high temperatures with random spin configuration, equilibrium configurations are obtained through cooling process. **Nanotube sizes studied:** 8×29, 12×44, 16×15



Hamiltonian: $\hat{H} = E_{ex} + E_{dip} + E_{ani}$, with key parameter: $\gamma = g/J$

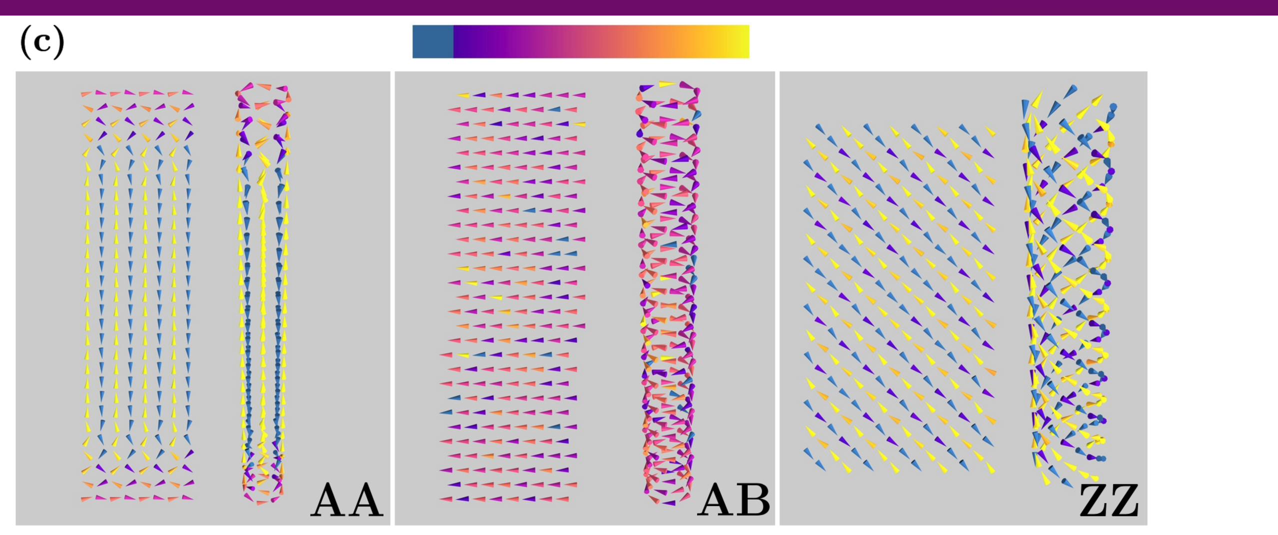
- **Exchange:** $E_{ex} = -\sum_{\langle i,j\rangle} J_{ij}\vec{S}_i \cdot \vec{S}_j$
- **Dipolar:** $E_{dip} = g\sum_{i<j} \frac{\vec{S}_i \cdot \vec{S}_j - 3(\vec{S}_i \cdot \hat{r}_{ij})(\vec{S}_j \cdot \hat{r}_{ij})}{|\vec{r}_{ij}|^3}$
- **Anisotropy:** $E_{ani} = -K\sum_i (\vec{S}_i \cdot \vec{n}_i)^2$

4. Pure Dipolar Results

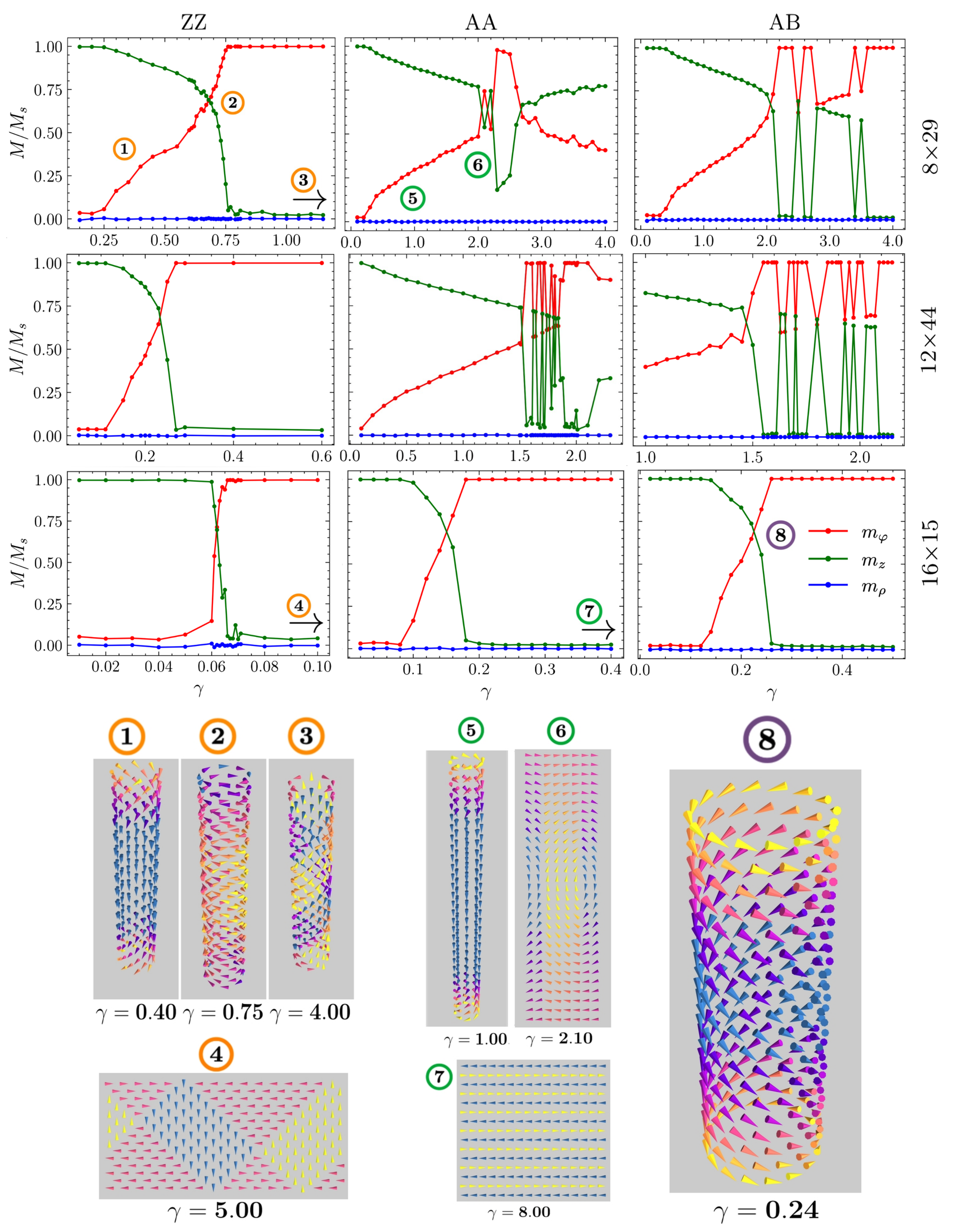
Pure dipolar interactions ($\gamma \to \infty$, $J = 0$):

- **AA**: FM along axis, AFM between columns
- **AB**: Vortex configuration, tangential spins
- **ZZ**: Antiferromagnetic helical order

The spatial arrangement of spins influences the magnetic order due to the long-range nature of dipolar interactions. Different lattice symmetries lead to distinct ground state configurations.



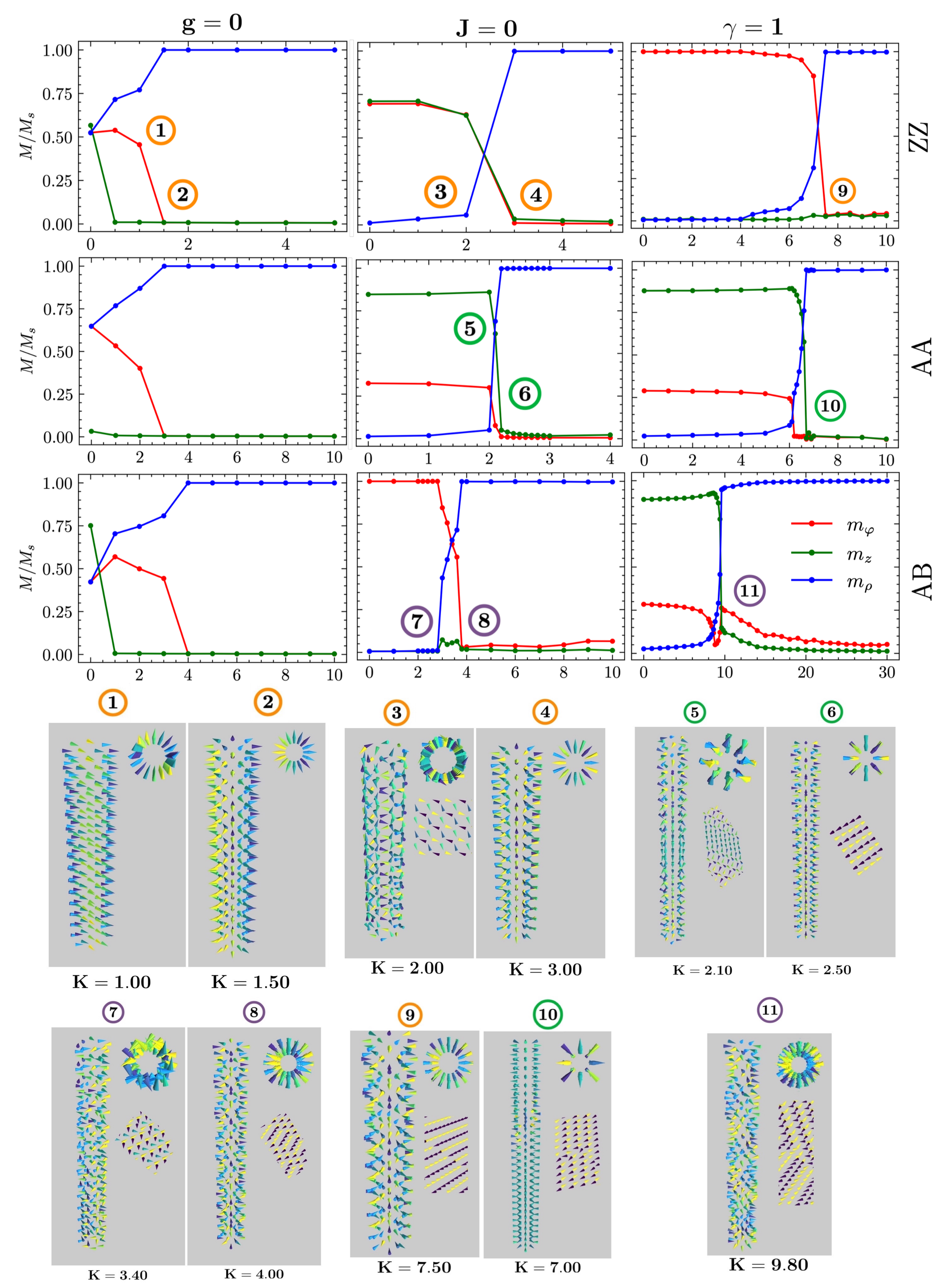
5. Competing Interactions



Exchange vs Dipolar competition: main transitions

- **ZZ Nanotubes**: FM → Vortex → AFM Helical ($\gamma_c \sim 0.7,\ 4.0$–8.0)
- **AA Nanotubes**: FM → V → AFM → V ($\gamma_c \sim 2.1,\ 8.0$)
- **AB Nanotubes**: FM → Vortex ($\gamma_c \sim 2$)

6. Adding anisotropy



Radial Anisotropy Effects: three scenarios studied

- **Exchange + Anisotropy** ($g = 0$): Isotropic FM → radial FM → flower states
- **Dipolar + Anisotropy** ($J = 0$): Geometry-dependent → Radial AFM states
- **All interactions** ($\gamma = 1$): Converge to radial AFM with tube-specific patterns

Critical points: ZZ ($K_c \sim 7.5$), AA ($K_c \sim 6.7$), AB ($K_c \sim 9.6$)

7. Conclusions

1. Competition results in rich phase diagrams with tube-specific configurations.
2. Geometry and chirality determine magnetic order and critical points.
3. Radial anisotropy stabilizes radial and mixed textures.
4. Results provide insight for functional magnetic nanostructure design.

8. Acknowledgements

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